UDACITY
Part of Accenture

Calculation of Basketball Scoring Probabilities

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Agenda

Results of basketball scoring calculations

- Overall Shooting Statistics
- Probabilities
- Conditional Probabilities - Future
- Conditional Probabilities - Retrospective

Overall Shooting Statistics

Section 1 — Formula

Empirical (frequentist) probability — estimated directly from data:

$$P(A) = \frac{\text{count}(A)}{N}$$

where N is the total number of observations.

Applied here:

$$P(\text{make}) = \frac{\text{count}(\text{result} = \text{True})}{N}$$

$$P(\text{miss}) = 1 - P(\text{make})$$

$$P(\text{3-pointer}) = \frac{\text{count}(\text{shot_type} = 3)}{N}$$

$$P(\text{2-pointer}) = \frac{\text{count}(\text{shot_type} = 2)}{N}$$

Section 1 - Overall Shooting Statistics

- Overall probability of a make (result = TRUE)
- Overall probability of a miss (result = FALSE)
- Overall proportion of three-pointers (shot_type = 3)
- Overall proportion of two-pointers (shot_type = 2)

Overall Shooting Statistics

```
==========================================
Section 1: OVERALL SHOOTING STATISTICS
==========================================
Metric                          Probability
------------------------------------------
P(Make)                              0.4902
P(Miss)                              0.5098
P(Three-pointer)                     0.5481
P(Two-pointer)                       0.4519
==========================================
```

Probabilities

Section 2 — Formula

Binomial distribution — probability of exactly k successes in n independent trials:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Symbol	Meaning
n	number of trials (shots)
k	desired number of successes
p	probability of success per trial

Assumptions: shots are independent and p is constant across shots.

Probabilities

Section 2 — Probabilities

Questions to Answer

1. What is the probability of Steph **making 3 of the next 4 shots**?
2. What is the probability that **4 of the next 5 shots are three-pointers**?

```python
from scipy.stats import binom

# 1.What is the probability of making 3 of the next 4 shots?
# p_q1 = P(3 makes in 4 shots)
p = p_make                      # probability of success (make)
n_shots = 4                     # number of trials
n = n_shots
k_successes = 3                 # number of successes
k = k_successes
p_q1 = binom.pmf(k, n, p)       # probability of exactly k successes in n trials
print(f"Probability of making exactly 3 out of the next 4 shots: {p_q1:.4f}")
```

```
Probability of making exactly 3 out of the next 4 shots: 0.2402
```

Probabilities

Applied numerically:

Q1 — P(exactly 3 makes in 4 shots), with $p = 0.4902$:

$$P(X = 3) = \binom{4}{3} \times 0.4902^3 \times 0.5098^1 = 4 \times 0.1177 \times 0.5098 \approx 0.2402$$

Q2 — P(exactly 4 three-pointers in 5 shots), with $p = 0.5481$:

$$P(X = 4) = \binom{5}{4} \times 0.5481^4 \times 0.4519^1 = 5 \times 0.0902 \times 0.4519 \approx 0.2039$$

Probabilities

```
============================================================
                SECTION 2 — PROBABILITIES
============================================================
Question                                        Probability
------------------------------------------------------------
P(exactly 3 makes in 4 shots)                     0.2402
P(exactly 4 three-pointers in 5 shots)            0.2039
============================================================

Assumptions:
  - Shots are independent events
  - p(make) and p(three) are constant across shots
  - p(make)  = 0.4902 (estimated from data)
  - p(three) = 0.5481 (estimated from data)
============================================================
```

Conditional Probabilities - Future

Section 3 — Formula

Conditional probability — probability of A given that B already occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{count}(A \text{ and } B)}{\text{count}(B)}$$

The joint probability is calculated as:

$$P(A \cap B) = \frac{\text{count}(A \text{ and } B)}{N}$$

Key rule: the notation $P(A \mid B)$ means:

- **Left of** $|$ → what you want to discover
- **Right of** $|$ → evidence you already have

Conditional Probabilities - Future

Applied in this section:

$$P(\text{make} \mid \text{three}) = \frac{P(\text{make} \cap \text{three})}{P(\text{three})}$$

$$P(\text{lead} \mid \text{three}) = \frac{P(\text{lead} \cap \text{three})}{P(\text{three})}$$

$$P(\text{make} \mid \text{two}) = \frac{P(\text{make} \cap \text{two})}{P(\text{two})}$$

$$P(\text{lead} \mid \text{two}) = \frac{P(\text{lead} \cap \text{two})}{P(\text{two})}$$

Conditional Probabilities - Future

Ex1: What is the probability of making a shot given that it is a three-pointer?

```
# Section 3: Conditional Probabilities
# Ex1: What is the probability of making a shot given that it is a three-pointer?
# P(make | three) = P(make ∩ three) / P(three)
# Step 1: P(make ∩ three) — joint probability
count_make_and_three = df[(df['result']) & (df['shot_type'] == 3)].shape[0]
p_make_and_three = count_make_and_three / total_shots
```

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```
# Step 2: Apply conditional probability formula
p_make_given_three = p_make_and_three / p_three

print(f"Count(make AND three): {count_make_and_three}")
print(f"P(make ∩ three):        {p_make_and_three:.4f}")
print(f"P(three):               {p_three:.4f}")
print(f"P(make | three):        {p_make_given_three:.4f}")
```

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```
Count(make AND three): 329
P(make ∩ three):        0.2294
P(three):               0.5481
P(make | three):        0.4186
```

Conditional Probabilities - Future

```python
# Section 3: Conditional Probabilities
# Ex2: What is the probability of lead given the shot is three-pointer ?
# P(lead | three) = P(lead ∩ three) / P(three)
# Step 1: P(lead ∩ three ) - joint probability
count_lead_and_three = df[df['lead'] & (df['shot_type'] == 3)].shape[0]
p_lead_given_three = count_lead_and_three / count_shot_type[3]

print(f"P(lead | three): {p_lead_given_three:.4f}")
```

```
P(lead | three): 0.4936
```

```python
# Step 2: Apply conditional probability formula
p_lead_and_three = count_lead_and_three / total_shots

print(f"Count(lead AND three): {count_lead_and_three}")
print(f"P(lead ∩ three):        {p_lead_and_three:.4f}")
print(f"P(three):               {p_three:.4f}")
print(f"P(lead | three):        {p_lead_given_three:.4f}")
```

```
Count(lead AND three): 388
P(lead ∩ three):        0.2706
P(three):               0.5481
P(lead | three):        0.4936
```

Conditional Probabilities - Retrospective

Section 4 — Formula

Bayes' Theorem — inverts the conditional: you have $P(B \mid A)$ but want $P(A \mid B)$:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Term	Name	Meaning
$P(A)$	**Prior**	Belief about A before seeing evidence
$P(B \mid A)$	**Likelihood**	Probability of evidence given cause
$P(B)$	**Evidence**	Total probability of observing B
$P(A \mid B)$	**Posterior**	Updated belief after evidence

Applied in this section:

$$P(\text{three} \mid \text{make}) = \frac{P(\text{make} \mid \text{three}) \cdot P(\text{three})}{P(\text{make})}$$

$$P(\text{two} \mid \text{make}) = \frac{P(\text{make} \mid \text{two}) \cdot P(\text{two})}{P(\text{make})}$$

> **Why Bayes?** In Section 3 you had the evidence (shot type) and wanted the outcome. Here the direction is **reversed**: you know the outcome (make) and want to infer the shot type.

Conditional Probabilities - Retrospective

Section 4 — Conditional Probabilities (Past)

Questions to Answer

If Steph **just made a shot**:

1. What is the probability that it was a **three-pointer**?
2. What is the probability that it was a **two-pointer**?

> **Note:** The project says you *could* answer these by direct counting, but you **must** use Bayes' Theorem and show your steps.

Conditional Probabilities - Retrospective

```python
# Section 4: Conditional Probabilities (Past) — Bayes' Theorem
# P(three | make) = P(make | three) * P(three) / P(make)
# P(two   | make) = P(make | two)   * P(two)   / P(make)

# Step 1: P(make | two) — likelihood needed for Ex2 (not yet calculated)
count_make_and_two = df[df['result'] & (df['shot_type'] == 2)].shape[0]
p_make_and_two = count_make_and_two / total_shots
p_make_given_two = p_make_and_two / p_two

print(f"Count(make AND two):  {count_make_and_two}")
print(f"P(make | two):        {p_make_given_two:.4f}")
print(f"P(make | three):      {p_make_given_three:.4f}  (from Section 3)")
```

```
Count(make AND two):  374
P(make | two):        0.5772
P(make | three):      0.4186  (from Section 3)
```

Conditional Probabilities - Retrospective

```python
# Step 2: Apply Bayes' Theorem
# Ex1: P(three | make) = P(make | three) * P(three) / P(make)
p_three_given_make = (p_make_given_three * p_three) / p_make

# Ex2: P(two | make) = P(make | two) * P(two) / P(make)
p_two_given_make = (p_make_given_two * p_two) / p_make

print(f"P(three | make): {p_three_given_make:.4f}")
print(f"P(two   | make): {p_two_given_make:.4f}")
```

```
P(three | make): 0.4680
P(two   | make): 0.5320
```

Conditional Probabilities - Retrospective

```
============================================================
        SECTION 4 — CONDITIONAL PROBABILITIES (PAST)
============================================================

Bayes Theorem components
  P(make | three) = 0.4186  (likelihood)
  P(make | two)   = 0.5772  (likelihood)
  P(three)        = 0.5481  (prior)
  P(two)          = 0.4519  (prior)
  P(make)         = 0.4902  (evidence)

Question                                  Probability
-------------------------------------------------------
P(three | make)                               0.4680
P(two   | make)                               0.5320
============================================================

Sanity check — P(three|make) + P(two|make) = 1.0000 (must be 1.0)
```

Conditional Probabilities - Retrospective

```python
# P(make | two) = P(make ∩ two) / P(two)
count_make_and_two = df[df['result'] & (df['shot_type'] == 2)].shape[0]
p_make_and_two = count_make_and_two / total_shots
p_make_given_two = p_make_and_two / p_two

print(f"Count(make AND two): {count_make_and_two}")
print(f"P(make ∩ two):       {p_make_and_two:.4f}")
print(f"P(two):              {p_two:.4f}")
print(f"P(make | two):       {p_make_given_two:.4f}")
```

```
Count(make AND two): 374
P(make ∩ two):       0.2608
P(two):              0.4519
P(make | two):       0.5772
```

Conditional Probabilities - Retrospective

Ex3 — Retrospective: P(lead | make)

> **Retrospective probability:** we already know the outcome (Steph *made* the shot) and are inferring the context (was the team leading?) — reasoning backward from result to circumstance.

Terms needed:

Term	Name	Value
$P(\text{lead})$	Prior	proportion of shots taken while leading
$P(\text{make} \mid \text{lead})$	Likelihood	make rate when team was leading
$P(\text{make})$	Evidence	overall make rate

Via Bayes' Theorem:

$$P(\text{lead} \mid \text{make}) = \frac{P(\text{make} \mid \text{lead}) \cdot P(\text{lead})}{P(\text{make})}$$

Equivalently, via direct joint probability (both give the same result):

$$P(\text{lead} \mid \text{make}) = \frac{P(\text{make} \cap \text{lead})}{P(\text{make})} = \frac{\text{count}(\text{make AND lead})}{\text{count}(\text{make})}$$

Conditional Probabilities - Retrospective

```
Bayes components:
  P(lead):              0.5035  (prior)
  P(make | lead):       0.5471  (likelihood)
  P(make):              0.4902  (evidence)
  P(make ∩ lead):       0.2755  (joint)

P(lead | make) via Bayes:        0.5619
P(lead | make) via direct joint:0.5619
Sanity check — both match: True
```

Conditional Probabilities - Retrospective

```python
# P(lead | two) = P(lead ∩ two) / P(two)
count_lead_and_two = df[df['lead'] & (df['shot_type'] == 2)].shape[0]
p_lead_and_two = count_lead_and_two / total_shots
p_lead_given_two = p_lead_and_two / p_two

print(f"Count(lead AND two): {count_lead_and_two}")
print(f"P(lead ∩ two):       {p_lead_and_two:.4f}")
print(f"P(two):              {p_two:.4f}")
print(f"P(lead | two):       {p_lead_given_two:.4f}")
```

```
Count(lead AND two): 334
P(lead ∩ two):       0.2329
P(two):              0.4519
P(lead | two):       0.5154
```

Conditional Probabilities - Retrospective

```
Count(make AND two): 374
P(make ∩ two):       0.2608
P(two):              0.4519
P(make | two):       0.5772
```

```
Count(lead AND two): 334
P(lead ∩ two):       0.2329
P(two):              0.4519
P(lead | two):       0.5154
```

Conditional Probabilities - Retrospective

Section 4 — Conclusions

- **P(Three | Make) = 0.4680** — when Steph makes a shot, there is a ~47% chance it was a three-pointer.
- **P(Two | Make) = 0.5320** — when Steph makes a shot, there is a ~53% chance it was a two-pointer.
- **P(Lead | Make) = 0.5619** — when Steph makes a shot, there is a ~56% chance his team was leading at the time.

These values match exactly the conditional probabilities calculated as intermediate terms in Section 3, which is a strong internal consistency check confirming that all results are coherent across sections.

Even though Steph attempts more three-pointers — P(Three) = 0.5481 — made shots are slightly more likely to be two-pointers. The reason is that his two-point accuracy is meaningfully higher: P(Make | Two) = 0.5772 versus P(Make | Three) = 0.4186. The higher conversion rate on two-pointers offsets the lower attempt volume, shifting the posterior toward two-pointers given a make.